

# Quantum State Discrimination

$$X \rightarrow \rho = \sum_{k=0}^{m-1} p_k \rho_k$$

$\rho_a$  is chosen at random with probability  $p_a$

$X$  is prepared in the state  $\rho_a$

we have to predict the value of  $a$   
by measuring  $X$



$$P_0 = \sum_{k \in S_0} |4_k\rangle\langle 4_k|$$

$$P_1 = \sum_{k \in S_1} |4_k\rangle\langle 4_k|$$

we perform measurement using  $\{P_0, P_1\}$

→ Correctness Probability

$$P_c = p \cdot \text{Tr}(P_0 \rho_0) + (1-p) \text{Tr}(P_1 \rho_1)$$

$$P_0 + P_1 = I \rightarrow P_1 = I - P_0$$

$$P_c = \text{Tr}(P_0 (p \rho_0 - (1-p) \rho_1))$$

$$\text{Tr}(\rho_1) = 1$$

$$+ (1-p) - \textcircled{1}$$

$$P_0 = I - P_1 \quad \text{Tr}(\rho_0) = 1$$

$$P_c = p - \text{Tr}(P_1 (p \rho_0 - (1-p) \rho_1)) - \textcircled{2}$$

$$P_c = \frac{\textcircled{1} + \textcircled{2}}{2} = \frac{1}{2} (1 + \text{Tr}((P_0 - P_1)(P \rho_0 - (1-p) \rho_1)))$$

$$(P_0 - P_1)(P \rho_0 + (1-p) \rho_1) = P_0(P \rho_0 - (1-p) \rho_1) - P_1(P \rho_0 - (1-p) \rho_1)$$

$$= \sum_{k \in S_0} \lambda_k |\psi_k\rangle\langle\psi_k| + \sum_{k \in S_0} -\lambda_k |\psi_k\rangle\langle\psi_k|$$

$$(P_0 - P_1)(P \rho_0 + (1-p) \rho_1) = \sum_{k=0}^{n-1} |\lambda_k| |\psi_k\rangle\langle\psi_k|$$

$$\text{Tr}((P_0 - P_1)(P \rho_0 - (1-p) \rho_1)) = \sum_{k=0}^{n-1} |\lambda_k|$$



## Discriminating between three or more states

→ No closed-form solution

→ Semidefinite program allows approximations

→ Holevo-Yuen-Kennedy-Lax condition

→ It is however, possible to verify/falsify optimality using H.Y.K.L. condition

$\{p_0, \dots, p_{m-1}\}$  is optimal if and only if

$$Q_a = \sum_{b=0}^{m-1} p_b \rho_b \rho_b - p_a \rho_a$$

follows  $Q_a \geq 0$  for all  $a \in \{0, \dots, m-1\}$

